Data Engineering - Course Project

Artificial Intelligence degree at USJ

Project

Introduction

This course project is an essential part for the calculation of the final mark of the subject, standing for 30% of the score. The project is very open to the student decision in most cases but should follow the guidelines stated in this document. The statement of the problem to solve is simple:

Design and test own data engineering solution with all things learned on this course.

This should be aligned with the next section's requirements.

Requirements

The project deliverable should meet the following requirements to be considered for scoring:

- The delivery should be made to a GitHub repository, where all the code, documentation and figures you want to show must be placed.
- The code should be able to run end-to-end and must be commented on. The main programming language used must be python, but other languages also can be considered. Best programming practices must be followed.
- The student must choose at least two datasets to work with, and a maximum of five. Election of these datasets should be justified and must be relatable.
- The project should cover all parts of the data life cycle, with processes to manage each one.
- All processes should be orchestrated to run at least once a day. As we are not performing in a production environment, it may not be feasible if we put it in our local machine, but at least a configuration (and explanation) should be provided on that aspect.
- Monitorization should be a must. That is, to design and implement a logging and alerting.
- At least an ETL process should be designed and implemented: read the datasets, integrate them, and put it on a well modelled data warehouse.
- Use of cloud services will be worth considering, but it should be used from python APIs.
- Data insights into the processed data should be taken. A free Looker Studio dashboard would be to consider for this.

- Documentation should account for all projects, including flow diagrams.
- The scalability of the project should be considered, so you need to infer the performance and costs of multiplying the amount of raw data by x10, x100, x1000, x10^6 (That is, the number of rows). A proposal should be made to address those problems.
- How much money will it cost if we migrate this to a cloud provider? Consider the x10, x100, x1000, x10^6 scenarios. Take one cloud provider to calculate it.
- Think about who can consume the data and design a solution to deliver it.
- Explain how AI can help with any of the processes.
- Is there any concern about privacy?
- English must be used always!

Timeline

Projects should be delivered before the 31st of December of 2025. Any update to the GitHub repository after this date will not be considered.

Mark calculation

The project mark will be created using the rubric in the appendix as guide. Content will be 90%, and language will be 10%. Language part for the project deliverable will be directly the full mark, unless there are words in Spanish or some clear error on an explanation due to the language.

Oral Presentation

Oral presentation of the project will be necessary to get the Oral Presentation as part of the mark, as the study guide of this course shows. Here the language will be graded 50%-50% with the content following the rubric in the appendix.

Timeline

The dates for the oral presentation will be defined from the second half of November to the first half of December. As you see, you don't have to have ended the project, but you need to present the solution you designed to a, not experienced, but beginner into the field audience.

Duration

The duration of the presentation will be up to 20 minutes, giving 10 minutes to answer questions. Making questions your classmates' presentations will be considered when calculating your mark so please participate!

Project Assessment Rubrics

Content rubric

Criteria	Excellent (3 points)	Development (2 points)	Needs Improvement (1 point)
Technical Accuracy & Depth	The explanation of the data engineering concept (e.g., its components, process flow) is accurate, specific, and demonstrates a deep understanding of the underlying principles.	The explanation is generally accurate but may oversimplify key components or contain minor technical inaccuracies.	The explanation contains significant technical errors or fundamentally misunderstands the concept.
Clarity & Use of Analogy	Complex technical ideas are made exceptionally clear for a non-expert audience through effective, relevant analogies (e.g., comparing a data pipeline to a factory assembly line).	An attempt is made to simplify the topic, but it may still rely on some unexplained technical jargon or use confusing analogies.	The explanation is abstract, filled with unexplained jargon, and makes no effective attempt to connect with a non-technical mindset.
Business Impact & Application	The "why it matters" is clearly translated into a tangible business outcome (e.g., cost savings, faster insights, new product features). The real-world application is concrete and highly relevant.	A business benefit is mentioned, but the link between the technical concept and the business impact is generic or not fully explained. The application is plausible but lacks detail.	The explanation fails to connect the technical work to any clear business value or the suggested application is irrelevant.

Language (communication skills) rubric

Vocabulary & Terminology	Key data engineering terms (e.g., "pipeline," "schema," "batch processing") are used correctly AND are immediately defined or simplified using clear, non-technical language.	Key terms are used, but their explanations are sometimes omitted, or still too technical for the intended audience.	The presentation is overloaded with unexplained acronyms (ETL, DAG, CI/CD) and technical terms, making it inaccessible.

Structure & Organization	The explanation follows a clear and logical narrative (e.g., Business Problem → Proposed Technical Solution → How it Works (Simplified) → Business Impact). The brief and video are easy to follow.	The information has a basic structure, but the flow between the business problem and the technical details feels disjointed or hard to follow at times.	The information is disorganized, jumping between technical and business points without a clear logical thread.
Register, Tone & Delivery	The tone is professional, confident, and accessible, effectively bridging the gap between a technical expert and a business stakeholder. The video delivery is engaging and clear.	The tone is generally appropriate but may occasionally become too academic or overly simplistic. The delivery is mostly clear but lacks engagement.	The tone is inappropriate for the audience (e.g., condescending or overly technical). The video delivery is unclear, rushed, or unengaging.